ZIHAO LU

SOFTWARE ENGINEER / COMPUTER VISION RESEARCHER

Würzburg, Germany
zihao.lu.henry96@gmail.com

Homepage Google Scholar GitHub LinkedIn

Computer vision research. Enterprise software experience.

I combine **four years of enterprise backend engineering at ICBC** with **computer vision research at the University of Würzburg**. My work spans RAW/RGB image restoration, learned image signal processing, and language-guided photo editing.

BMVC 2026

First author

RAW/RGB restoration

ICCV 2025

Second author

Language-guided editing

200k+

Transactions per day

Main systems (e.g., Visa)

PROFESSIONAL EXPERIENCE

Computer Vision Lab · University of Würzburg

Student Research Assistant (HiWi) | Oct 2023 - Sep 2026

- **First-author BMVC 2026 study:** curated data, designed degradations, trained models, and benchmarked RAW/RGB restoration across **four smartphone device groups, two learned ISPs, and three degradation settings**.
- **PixTalk, ICCV 2025:** contributed dataset construction, model training, prompt design, evaluation experiments, and manuscript preparation for language-guided photo editing.
- **NTIRE 2024/2025:** supported data preparation and experiment reproduction; co-organized the NTIRE 2025 RAW Image Restoration and Super-Resolution Challenge.

ICBC · Software Development Center

Senior Backend Developer | Jul 2018 - Jun 2022

- Core developer and on-call engineer for **six large-scale systems and three regional subsystems**, operating under a **99.99% reliability SLA**.
- Supported **American Express** processing in **Hong Kong and Macau** (regional subsystem: approximately **10,000 transactions/day**) and main card-payment systems such as **Visa** (**200,000+ transactions/day**).
- Led a **five-person team**, allocating development and testing tasks and communicating simplified development guidelines. Served as a key resource for business rules and system behavior.

RESEARCH DIRECTION

I study how AI can improve photographs from RAW sensor data to the final image, while giving people intuitive control over the result. My own photography informs this work.

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SELECTED PUBLICATIONS

Benchmarking RAW and RGB Restoration in Image Signal Processors

Zihao Lu, Radu Timofte, Marcos V. Conde

BMVC 2026 · Accepted · First author

Benchmarks restoration before and after a fixed ISP. Results highlight the importance of matching restoration training to the target imaging pipeline.

Paper Code

PixTalk: Controlling Photorealistic Image Processing and Editing with Language

Marcos V. Conde, **Zihao Lu**, Radu Timofte

ICCV 2025 · Second author

A vision-language model for controllable photo editing. The published method supports over 40 transformations and processes 12-megapixel images in under one second on consumer GPUs.

Paper Project

NTIRE 2025 Challenge on RAW Image Restoration and Super-Resolution

Marcos V. Conde, Radu Timofte, **Zihao Lu**, et al.

CVPR Workshops 2025 · Challenge co-organizer and co-author

Challenge report covering RAW restoration and 2x super-resolution, with 230 registered participants and 45 submitting results.

Paper Workshop

ACADEMIC SERVICE

ACCV 2026 · Imaging for Physical AI (2nd Edition) · Workshop co-organizer

CVPR 2025 / NTIRE · RAW challenge co-organizer; workshop organization support

CVPR 2024 / NTIRE · Workshop organization support

EDUCATION

M.Sc. Computer Science

Julius-Maximilians-Universität Würzburg

Nov 2022 - Aug 2026

B.Sc. Computer Science

Southwest Jiaotong University

Sep 2014 - Jul 2018

TECHNICAL TOOLKIT

ML / vision: PyTorch, NumPy, OpenCV, CUDA

Backend: Java, C, Oracle, MySQL, ZooKeeper

Tools: Linux, Docker, Git, Kubernetes

LANGUAGES

Chinese: Native **English:** B2 (IELTS 6.5) **Japanese:** JLPT N2 **German:** A1